Notes: Maksimovic & Phillips (Journal of Finance, 2002)
Do Conglomerate Firms Allocate Resources Inefficiently Across Industries? Theory and Evidence

Contents

1 Big picture

- **Question.** Do conglomerates allocate resources inefficiently across business segments, or can observed conglomerate discounts and segment investment patterns arise from profit-maximizing behavior?
- **Main challenge to the literature.** Earlier conglomerate-discount and internal-capital-market papers often use single-segment industry Tobin's q as the benchmark for a conglomerate segment's investment opportunities. Maksimovic and Phillips argue that this benchmark is potentially misspecified because firms in the same industry may have different productivity and organizational capabilities.
- **Core mechanism.** Firms differ in industry-specific organizational ability. A firm may optimally diversify if its relative abilities across industries are close enough. Its largest segments should generally be its most productive segments, and its growth across segments should respond to relative productivity and industry demand shocks.
- **Data advantage.** The paper uses plant-level U.S. Census Longitudinal Research Database data, rather than reported Compustat segment data, to construct firm-by-industry segments and measure productivity directly at the plant and segment level.
- **Main empirical result.** The majority of conglomerates allocate resources in a way consistent with a neoclassical profit-maximizing model: productive segments grow more, and a segment grows less when other segments have better opportunities.
- **Conglomerate discount interpretation.** A conglomerate discount can be endogenous: diversified firms may be less productive than focused firms of comparable size, especially because peripheral segments have lower productivity, without this necessarily implying inefficient internal capital markets.
- **Agency evidence.** There is some evidence that conglomerates that later restructure do not follow the model as well, but even for these firms the paper does not find strong evidence that less productive segments are subsidized.
- **Economic takeaway.** Diversification and internal resource allocation should be judged relative to a productivity-based benchmark, not simply relative to single-segment firms' industry q. Once firm-specific productivity is measured, much conglomerate behavior looks closer to efficient comparative-advantage allocation than to indiscriminate cross-subsidization.

2 Incremental contribution of the paper

- **A productivity-based benchmark for conglomerate investment.** The paper's most important contribution is to replace the common "same industry, same opportunities" assumption with a benchmark in which firms have different ability within the same industry. This changes how one interprets the conglomerate discount and segment investment regressions.
- **Linking theory and plant-level evidence.** The paper derives predictions from a neoclassical model of firm size and diversification and then tests those predictions using microdata. The model predicts not merely that productive segments grow, but also that growth in one segment should depend negatively on opportunities in the firm's other segments.
- **Direct measurement of productivity.** By using plant-level Census data, the authors observe production, inputs, and plant ownership. This permits a segment-level productivity measure that is closer to production fundamentals than market-based industry q.
- **New interpretation of the conglomerate discount.** The paper shows that lower average productivity among conglomerates can be consistent with optimizing behavior and decreasing returns to organizational ability. A discount does not automatically imply misallocation.
- **Sharper test of cross-subsidization.** Agency stories often imply that cash or good shocks in one segment subsidize weak segments. The paper tests a different implication: if other segments have better

productivity and better demand conditions, a segment should grow less under efficient allocation. The evidence supports this prediction for most conglomerates.

- **Distinguishing surviving from broken-up conglomerates.** The paper introduces a useful robustness split. Conglomerates that survive look much more consistent with the model, while those that later restructure fit the model less well. This separates a broad efficient-allocation pattern from a smaller set of potentially problematic conglomerates.

3 Data construction and measurement

3.1 Plant-level data from the Census Longitudinal Research Database

- The paper uses the Longitudinal Research Database (LRD) maintained by the U.S. Census Bureau.
- The LRD contains detailed plant-level data on value of shipments, investment in equipment and buildings, employment, and inputs.
- The sample covers manufacturing plants in SIC 2000–3999.
- The underlying data span 1974–1992. Because productivity is estimated using five-year plant histories, the main productivity-based regressions use 1979–1992.
- The sample includes approximately 767,098 plant-year observations after requiring at least two years of plant data and excluding small firm-industry cells with less than one million real dollars of shipments.

Interpretation: The plant-level design is central. The paper is not relying on firm-level market valuation or manager-reported segment boundaries. Instead, it rebuilds each firm's industry profile from plants and SIC codes, which makes it possible to compare real production efficiency across a firm's segments.

3.2 Constructing firm segments from plants

- A firm segment is a group of plants owned by the same firm in the same three-digit SIC industry.
- Single-segment firms are firms operating in one three-digit SIC code.
- Multiple-segment conglomerates are firms operating in more than one three-digit SIC code.
- A conglomerate's **main segment** is its largest industry segment, defined by shipments.
- **Peripheral segments** are smaller segments. In some analyses, the paper requires segments to represent at least 2.5 percent or 10 percent of the firm's total shipments.

Interpretation: The constructed segments may not exactly match internal divisions, but they avoid the discretion embedded in reported segment data. The price is that the analysis focuses on manufacturing and three-digit SIC industry definitions.

3.3 Measuring productivity with a translog production function

For each industry, the paper estimates a translog production function using plant-level data:

$$\ln Q_{it} = A + f_i + \sum_{j=1}^{N} c_j \ln L_{jit} + \sum_{j=1}^{N} \sum_{k=j}^{N} c_{jk} \ln L_{jit} \ln L_{kit}, \tag{1}$$

where Q_{it} is output of plant i in year t, L_{jit} is input j, A is an industry technology parameter, and f_i is a plant-firm fixed effect.

- Plant productivity is the residual plus the fixed effect from the production-function regression.
- This productivity measure captures both technical efficiency and pricing ability relative to industry averages.
- Productivity is standardized by the standard deviation of productivity in the industry.
- Segment productivity is the predicted-output-weighted average of plant productivities within the firm-industry segment.

Interpretation: The fixed effect matters because persistent plant-level productivity may reflect managerial ability or organizational capital. Standardization avoids letting industries with more dispersed productivity dominate cross-industry comparisons.

3.4 Industry demand and business-cycle variables

- Real industry shipments are detrended using a yearly time trend.
- Expansion years are years in which both real and detrended aggregate industrial production increase relative to the prior year.
- Recession years are years in which both real and detrended aggregate industrial production decline relative to the prior year.
- Segment growth is industry adjusted each year, so the dependent variable captures growth relative to industry-year conditions.

Interpretation: The demand variables are designed to distinguish whether a segment is growing because its whole industry is booming or because the firm has a comparative advantage in that industry.

3.5 Key outcomes

- **Segment growth:** annual sales/shipment growth of a firm-industry segment, adjusted by industry and year.
- **Segment capital expenditures:** capital expenditure scaled by beginning-of-period capital stock, adjusted by industry and year.
- **Productivity/efficiency:** baseline TFP, plus robustness measures such as value added per worker and operating margin.

Interpretation: Growth and capital expenditures are complementary. Growth shows changes in segment output; capital expenditures test whether real investment follows the same allocation logic.

4 Theory and empirical specifications

4.1 Firm production and diversification choice

Consider a firm j that can operate in industries A and B. The firm has industry-specific productivity d_{Aj} and d_{Bj}. A simplified version of the profit function is:

$$\pi_j = d_{Aj} p_A k_{Aj} + d_{Bj} p_B k_{Bj} - r_A k_{Aj} - r_B k_{Bj} - u\ell_{Aj}^2 - u\ell_{Bj}^2 - w(\ell_{Aj} + \ell_{Bj})^2. \quad (2)$$

The final term captures diseconomies of scale from scarce organizational talent: expansion in one segment raises the cost of managing the rest of the firm.

Define

$$v_i = d_i p_i - r_i, \tag{3}$$

where v_i is the net productivity-adjusted return in industry i. For a conglomerate producing in both industries, optimal outputs are:

$$k_A = \frac{(u+w)v_A - wv_B}{2u(u+2w)}, \tag{4}$$

$$k_B = \frac{(u+w)v_B - wv_A}{2u(u+2w)}. \tag{5}$$

Interpretation: The firm does not diversify simply because diversification is mechanically valuable. It diversifies only when both industries are close enough in net return. If one industry dominates, specialization is optimal.

4.2 Proposition 1: when diversification is optimal

Let $\theta = (u+w)/w$. Diversification is optimal when:

$$\theta v_B > v_A \quad \text{and} \quad \theta v_A > v_B. \tag{6}$$

- If $v_B > \theta v_A$, the firm specializes in industry B.
- If $v_A > \theta v_B$, the firm specializes in industry A.
- If returns are close enough, the firm operates in both industries.

Interpretation: The model implies that a conglomerate's largest segment should be more productive than its smaller segments, because a firm expands further where it has greater comparative advantage.

4.3 Growth response to demand shocks

The model predicts that segment growth depends on own productivity, own industry demand, other-segment productivity, and other-segment demand conditions. A simplified empirical specification is:

$$Growth_{ist} = \alpha + \beta_1 Demand_{st} + \beta_2 TFP_{ist} + \beta_3 (Demand_{st} \times TFP_{ist}) + \beta_4 OtherTFP_{it} + \beta_5 (RelDemand_{ist} \times OtherTFP_{it}) + X'_{ist} \tag{7}$$

- $Growth_{ist}$ is industry-adjusted segment growth for firm i in segment/industry s and year t.
- $Demand_{st}$ is detrended industry shipments.
- TFP_{ist} is segment productivity.
- $OtherTFP_{it}$ is a weighted average of the firm's other segments' productivity.
- $RelDemand_{ist}$ compares the segment's industry demand change with the firm's median segment demand change.

Interpretation: In an efficient allocation model, a segment grows more when it is productive and its industry demand is strong. But it grows less when the firm's other segments are more productive and have relatively strong demand. This latter prediction is the key test against simple cross-subsidization stories.

4.4 Investment specification

The capital expenditure specification mirrors the growth specification:

$$Capex_{ist} = \alpha + \beta_1 Demand_{st} + \beta_2 TFP_{ist} + \beta_3 (Demand_{st} \times TFP_{ist}) + \beta_4 OtherTFP_{it} + \beta_5 (RelDemand_{ist} \times OtherTFP_{it}) + X'_{ist}\gamma \tag{8}$$

Interpretation: If growth results are driven by efficient real reallocation, capital expenditures should display similar productivity and demand sensitivities. If results are only accounting artifacts, the investment regressions should be weaker or inconsistent.

4.5 Alternative efficiency and focus specifications

The paper checks robustness by replacing TFP with:

- value added per worker;
- operating margin.

It also tests whether diversification focus, measured by a Herfindahl index of segment sales shares, changes resource allocation:

$$Outcome_{ist} = \cdots + \delta_1 Herfindahl_{it} \times TFP_{ist} + \delta_2 Herfindahl_{it} \times OtherTFP_{it} + \epsilon_{ist}. \tag{9}$$

Interpretation: These checks ask whether the main conclusion is sensitive to the exact productivity measure or to the degree of diversification.

5 Identification and empirical strategy

5.1 Conceptual identification problem

The classic internal-capital-market question asks whether conglomerates move resources inefficiently across segments. The identification problem is that segment investment opportunities are not directly observable. If researchers proxy investment opportunities with single-segment industry q, they implicitly assume that all firms in the same industry face the same marginal returns.

Interpretation: Maksimovic and Phillips' identification strategy is to construct a better benchmark: segment-level productivity measured from the firm's own plants. This allows them to compare a conglomerate segment's actual growth to its own productivity and its own industry demand conditions.

5.2 Step 1: show that conglomerates differ from single-segment firms

- The paper first documents that conglomerate plants are generally less productive than single-segment plants of similar size, except among the smallest firms.
- This undermines the assumption that a conglomerate segment and a standalone firm in the same industry share the same opportunities.
- It also provides a neoclassical interpretation of the conglomerate discount: lower productivity can generate lower value even without agency problems.

Interpretation: This step does not prove efficient allocation, but it establishes that earlier tests may have used a flawed counterfactual.

5.3 Step 2: test within-conglomerate comparative advantage

- The model predicts that larger segments should be more productive than peripheral segments.
- Figure 3 and Table II test this by ranking segments within conglomerates and by comparing main and peripheral segments.

- The evidence shows that main segments are generally more productive than peripheral segments.

Interpretation: If conglomerates were simply empire-building into weak segments, one might still see weak peripherals. The stronger test is whether growth and investment discipline those weaker peripherals.

5.4 Step 3: use demand shocks to test resource allocation

- The empirical specifications interact demand changes with segment productivity.
- Efficient allocation predicts stronger growth in productive segments facing positive demand shocks.
- It also predicts that a segment grows less when the firm's other segments are more productive and face relatively stronger demand.
- Cross-subsidization stories often predict the opposite: good shocks elsewhere provide resources to expand weak segments.

Interpretation: This is the paper's most important identifying contrast. The model is not just saying "productive segments grow." It says that other-segment productivity and demand should shift resources away from the focal segment.

5.5 Step 4: compare surviving and restructuring conglomerates

- The paper identifies conglomerates that later restructure by reducing the number of segments by at least 25 percent.
- If these firms are more likely to have agency problems, the model should fit them less well.
- Table IX shows that the model's key interaction terms are weaker or insignificant for restructuring firms, but significant for conglomerates that do not restructure.

Interpretation: This split supports a nuanced conclusion. Some conglomerates may be inefficient, but the majority of surviving conglomerates behave consistently with efficient allocation.

5.6 Main threats and limits

- Productivity may include pricing power as well as technical efficiency.
- The analysis focuses on manufacturing, so it may not generalize to services or finance.
- The tests do not capture headquarters overhead, managerial perks, or waste after cash flow is generated.
- Segment definitions based on SIC codes may differ from internal managerial divisions.
- The paper measures allocation across plants and segments, not stock market valuation directly.

Interpretation: The identification strategy is powerful for testing productive allocation at the plant and segment level, but it cannot rule out all possible agency costs or all forms of waste.

6 Tables and figures

6.1 Figure 1: Optimal production with differing ability across industries

Read:

- The figure maps firms into ability space for industries A and B.
- Region II contains firms for which diversification is optimal.
- Regions I and III contain firms whose comparative advantage is so skewed that specialization is optimal.
- The boundaries are defined by $v_B = \theta v_A$ and $v_A = \theta v_B$.

Interpretation: The figure gives the central intuition: conglomerates need not be inefficient. They can be firms whose organizational ability is valuable in more than one industry.

Economic message: A diversified firm can be a profit-maximizing response to comparative advantage, not necessarily an agency-driven expansion beyond core competence.

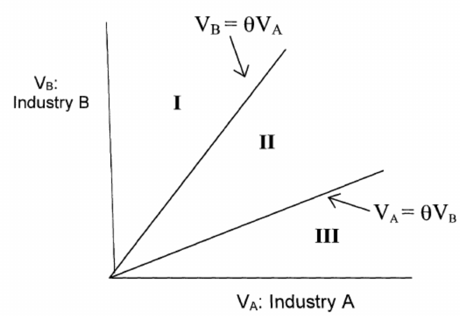
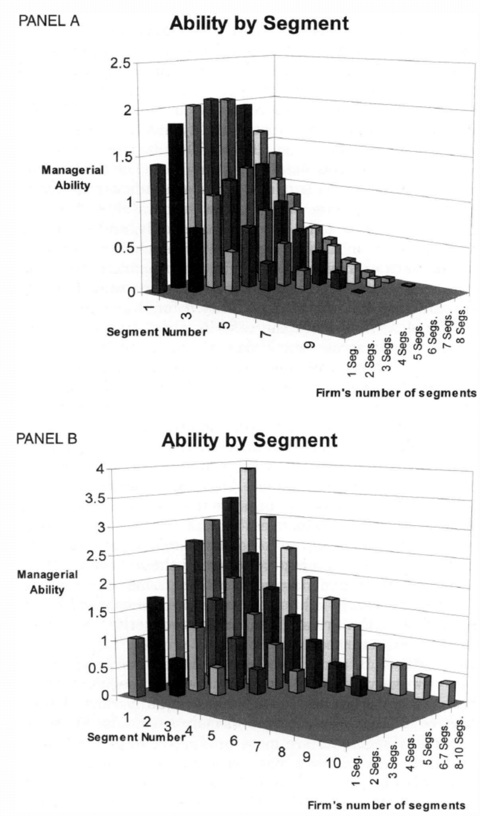


Figure 1. Optimal production with differing ability across industries.

Figure 1: Optimal production with differing ability across industries



'igure 2. Model with no common managerial ability across industries (Panel A) and

Figure 2: Model with no common managerial ability across industries (Panel A) and with managerial ability applicable across industries (Panel B)

6.2 Figure 2: Model with no common managerial ability across industries and with managerial ability applicable across industries

Read:

- Panel A simulates the case in which managerial ability is independent across industries.
- Main segments are more productive than peripheral segments.

- As the number of segments rises, equally ranked segments initially become more productive but later decline because it is unlikely that one firm is highly productive in many industries.
- Panel B adds a common firm-level ability component. Main segments remain more productive, but firms with more segments can have higher ability across equally ranked segments.

Interpretation: The figure shows that the productivity-focus relation depends on the distribution of ability. A conglomerate discount can arise even in a model with value-maximizing firms.

Economic message: The mere fact that conglomerates trade at a discount is not enough to infer inefficient internal capital markets. The discount may reflect equilibrium sorting by ability.

6.3 Table I: Sample characteristics - single-segment and multiple-segment firms

Read:

- In 1980, single-segment firms have 13,298 segments, while multi-segment firms have 4,880 main segments and 2,582 peripheral segments.
- Single-segment firms' share of manufacturing shipments rises from 21.71 percent in 1980 to 26.72 percent in 1990.
- Main segments account for roughly half of manufacturing shipments, while peripheral segments fall from 27.53 percent to 23.63 percent.
- Productivity rises with segment size for both single-segment and multi-segment firms.
- During expansion years, main segments of conglomerates grow faster than peripheral segments.

Interpretation: The table provides two key facts: size matters for productivity, and main segments are economically important. Peripheral segments are not trivial, but they are smaller and often less productive.

Economic message: The sample is consistent with the paper's comparative-advantage view: firms allocate more output to larger, more productive segments.

6.4 Figure 3: Productivity ordered by segment size

Read:

- The figure ranks segments within conglomerates and plots average segment TFP.
- First-ranked and larger segments are typically more productive than later-ranked segments.
- Productivity declines as segments become more peripheral.
- Firms with more segments can still have high productivity in leading segments.

Interpretation: The figure visually confirms a core prediction: conglomerates are not randomly spreading resources. Their largest segments tend to be the places where they have relatively high productivity.

Economic message: Peripheral weakness does not by itself prove misallocation; it may reflect optimal experimentation or limited comparative advantage.

Table I

Sample Characteristics: Single-segment and Multiple-segment firms

Sample characteristics of firms' industry operating segments. We calculate statistics from plant-level data aggregated into three-digit SIC codes for each firm. We classify single-segment versus multiple-segment firms based on three-digit SIC codes. For multiple-segment firms, main segments are segments that represent at least 25 percent of the firm's total shipments. We base size classifications on the previous year's real value of industry shipments relative to each industry's median value of shipments. We determine recession and expansion years using aggregate detrended industrial production. Recession years are years in which both real and detrended industrial production decline relative to the previous year. We classify years as expansion years when both real and detrended industrial production increase relative to the previous year.

Statistics by Industry Segments	Sample of Firms: Single-segment Firms	Multiple-segment Firms: Main Segments	Multiple-segment Firms: Peripheral Segments
Number of segments—beginning of decade: 1980	13,298	4,880	2,582
Number of segments—end of decade: 1990	17,321	4,745	3,090
Proportion of value of shipments (all manufacturing industries)			
Beginning of decade: 1980	21.71%	50.75%	27.53%
End of decade: 1990	26.72%	49.65%	23.63%
		Multiple-segment Firms	
Productivity by segment size (average over years in panel)[a]			
\$1–\$10 million in value of shipments, (1982 \$)	−0.417	−0.331	
\$10–\$25 million in value of shipments, (1982 \$)	0.148	0.109	
\$25–\$50 million in value of shipments, (1982 \$)	0.322	0.317	
\$50–\$100 million in value of shipments, (1982 \$)	0.404	0.383	
Segments > \$100 million (1982 \$)	0.487	0.449	
		Main Segments	Peripheral Segments
Average annual industry segment growth rate			
Recession years (1981–1982, 1990–1991)			
All industry segments	−5.15%	−0.01%	−5.48%
Firms' most productive segments (top 50th percentile of TFP by industry)	−3.54%	1.57%	−4.06%
Expansion years (1976–1978, 1984–1988)			
All industry segments	2.46%	7.30%	2.60%
Firms' most productive segments (top 50th percentile of TFP by industry)	5.99%	9.66%	4.35%

[a] Productivity is total factor productivity (TFP) and is a relative measure of productivity. TFPs are standardized by dividing each TFP by the standard deviation of the industry's TFP at the

Table I: Sample characteristics - single-segment and multiple-segment firms

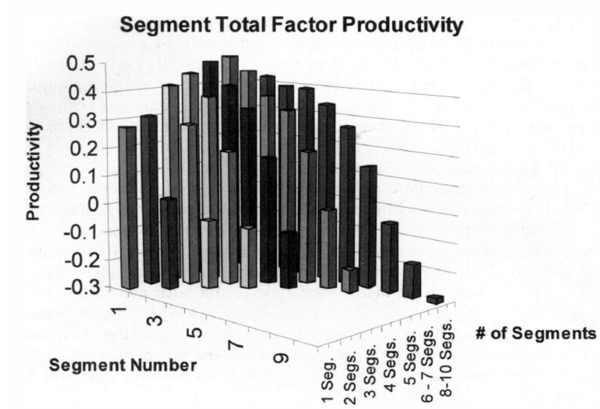


Figure 3. Productivity ordered by segment size.

Figure 2: Model with no common managerial ability across industries (Panel A) and with managerial ability applicable across industries (Panel B)

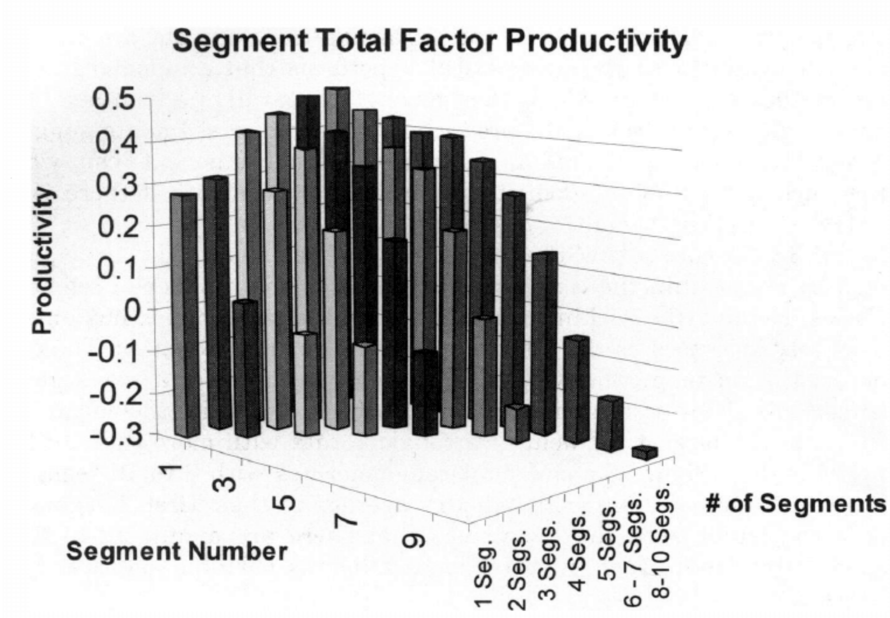


Figure 3. Productivity ordered by segment size.

Figure 3: Productivity ordered by segment size

6.5 Table II: Growth and productivity over the business cycle

Read:

- In expansions, efficient segments grow faster than less efficient segments across both single-segment and multi-segment firms.
- Main segments generally have higher productivity than peripheral segments across size groups.
- In recessions, growth is weaker, but productivity patterns remain informative.
- Peripheral segments are often less productive, especially in smaller size groups.

Interpretation: The table shows that segment growth is not random. Productivity and business-cycle conditions jointly shape growth patterns. Main segments tend to perform better because they are more productive, not merely because headquarters favors them.

Economic message: The business-cycle split matters because optimal resource allocation should respond differently when industry demand expands or contracts.

6.6 Table III: Segment growth

Read:

- Segment productivity has a positive and highly significant effect on growth for both single-segment and multi-segment firms.
- The interaction between industry shipments and TFP is positive, indicating that productive segments grow more when industry demand rises.

Table II

Growth and Productivity over the Business Cycle

Sample characteristics of firms' industry operating segments in expansion and recession years. Statistics are calculated from plant-level data aggregated into three-digit SIC codes for each firm. Single-segment versus multiple-segment firms are based on three-digit SIC codes. For multiple-segment firms, main segments are segments which are at least 25 percent of the firm's total shipments. Size classifications are based on previous years real value of industry shipments and are relative to each industry's median value of shipments. Number of segments is for the beginning of the period. Expansion (recession) years are years in which both real and detrended aggregate industrial production increase (decrease) relative to the previous year.

	Sample of Firms		
		Multiple-segment Firms	
	Single-segment Firms	Main Segments	Peripheral Segments
Panel A: Characteristics of Firm Operating Segments in Expansion Years (Average over 1976–1978, 1984–1988)			
Size Group 1: one-quarter to one-half of the industry median			
Real growth of firm segment	5.10%	9.44%	2.97%
Growth of efficient segments (top 50% of TFP)[a]	13.70%	17.20%	11.40%
Productivity (TFP) of firm segment[a]	−0.200	−0.121	−0.391[b]
Number of firm segments	2025	213	65
Size Group 3: one to two times the industry median			
Real growth of firm segment	2.13%	9.08%	2.30%
Growth of efficient segments (top 50% of TFP)[a]	7.27%	13.56%	7.32%
Productivity (TFP) of firm segment[a]	0.197[c]	0.120	−0.173[b]
Number of firm segments	1195	412	274
Size Group 5: greater than five times the industry median			
Real growth of firm segment	1.10%	6.15%	4.23%
Growth of efficient segments (top 50% of TFP)[a]	2.67%	7.49%	6.70%
Productivity (TFP) of firm segment[a]	0.422[c]	0.375	0.154[b]
Number of firm segments	171	731	3241
Panel B: Characteristics of Firm Operating Segments in Recession Years (1981–1982, 1990–1991 Averages)			
Size Group 1: segment one-quarter to one-half of the industry median			
Real growth of firm segment	−1.50%	2.16%	−2.30%
Growth of efficient segments (top 50% of TFP)	7.57%	7.31%	4.79%
Productivity (TFP) of firm segment	−0.272	−0.169	−0.582[b]
Number of firm segments	3380	116	31
Size Group 3: segment one to two times the industry median			
Real growth of firm segment	−5.10%	−0.75%	−8.10%
Growth of efficient segments (top 50% of TFP)	−0.03%	4.77%	−3.11%
Productivity (TFP) of firm segment	0.173[c]	0.061	−0.198[b]
Number of firm segments	2495	341	130
Size Group 5: segment greater than five times the industry median			
Real growth of firm segment	−6.20%	−0.84%	−4.20%
Growth of efficient segments (top 50% of TFP)	−3.16%	0.60%	−0.78%
Productivity (TFP) of firm segment	0.379[c]	0.373	0.175[b]
Number of firm segments	1226	1305	3967

[a] Productivity is Total Factor Productivity (TFP) and is a relative measure of productivity. TFPs are standardized by dividing each TFP by the standard deviation of the industry's TFP at the three-digit level.

[b] Significant difference between main and peripheral segments at less than the five percent level using a two-tailed test for a difference from zero.

[c] Significantly different from multiple-segment firms at less than the five percent level using a two-tailed

Table II: Growth and productivity over the business cycle

Table III

Segment Growth

Regressions test the effects of plant-level productivity and industry-level demand on firm industry segment sales growth for single-segment and multiple-segment firms. The dependent variable, segment growth is industry adjusted in each year. Segment size and productivity are industry adjusted in each year. Real industry shipments is detrended by regressing industry shipments (in 1987 dollars) on a yearly time trend. In column two (three) multiple-segment firms are firms with at least two segments each with 2.5 percent (10 percent) of their total shipments. Data are aggregated into firm three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber–White. Data are yearly from 1979 to 1992. Observations are included for growth rates less than 100 percent and for firms whose real value of shipments is greater than one million dollars. (p-values are in parentheses.)

Dependent Variable: Segment Growth				
		Multiple-segment Firms		Test for Significant Diff.: Multiple-segment Interaction Variable (p-value)[d] Column 3–Column 1
Variable	Single-segment Firms	(2.5 percent cutoff)	(10 percent cutoff)	
Constant	0.024	−0.004	−0.005	
	(0.000)[a]	(0.004)[a]	(0.003)[a]	(0.000)[a]
Real industry shipments (detrended)	0.009	0.002	0.005	
	(0.186)	(0.717)	(0.462)	(0.715)
Firm segment productivity (TFP)[e]	0.069	0.054	0.054	
	(0.000)[a]	(0.000)[a]	(0.000)[a]	(0.000)[a]
Industry shipments (detrended) * TFP	0.013	0.022	0.019	
	(0.021)[b]	(0.002)[c]	(0.013)[b]	(0.106)
ln(lagged firm segment size)	−0.047	−0.002	−0.002	
	(0.002)	(0.339)	(0.376)	(0.242)
Number of plants owned by firm	−1.371	−0.028	−0.028	
(beginning of year, coeff * 100)	(0.000)[a]	(0.000)[a]	(0.000)[a]	(0.000)[a]
Firm multiple-segment variables				
Other segment's weighted TFP[f]		−0.008	−0.007	
		(0.000)[a]	(0.000)[a]	
Relative demand * other segments		−0.003	−0.004	
weighted TFP[g]		(0.051)[b]	(0.035)[b]	
Total industry-segment years	171,126	94,977	87,915	
Chi-squared statistic	8871.14	2417.17	2111.30	
Significance level (p-value)	<1%	<1%	<1%	

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

[d] Significance test for a multiple-segment dummy variable interacted with each independent variable in a regression with all firm segments.

[e] Total Factor Productivity (TFP) is calculated using a translog production function.

[f] Other segments' productivity is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.

[g] Relative industry demand is interacted with other segments' productivity and equals one

Table III: Segment growth

- Other segments' weighted TFP is negative for conglomerates, meaning a segment grows less when other segments are more productive.
- Relative demand interacted with other segments' weighted TFP is negative and significant for multi-segment firms.

Interpretation: This is the main regression evidence. Growth follows comparative advantage: productive segments grow, but the focal segment loses resources when other parts of the firm have stronger opportunities.

Economic message: The results are difficult to reconcile with simple cross-subsidization of weak segments. They are more consistent with reallocation toward high-return segments.

6.7 Table IV: Segment growth for main and peripheral segments

Read:

- Both main and peripheral segments have positive sensitivity to own TFP.
- Peripheral segments show especially strong productivity sensitivity.
- The interaction of relative demand and other segments' weighted TFP is negative for both main and peripheral segments, and significant for peripheral segments.
- The difference between main and peripheral coefficients is not always statistically sharp, but signs follow the model.

Interpretation: Peripheral segments are not insulated from market discipline. If anything, they appear more sensitive to productivity, which weakens the view that conglomerates systematically subsidize weak peripheral activities.

Economic message: Resource allocation inside conglomerates looks closer to comparative advantage than to blanket internal capital-market support for low-productivity divisions.

6.8 Table V: Economic significance of regression results

Read:

- Moving productivity from the 25th to the 75th percentile changes predicted growth substantially.
- Single-segment growth rises from -2.69 percent to 4.39 percent.
- Conglomerate main-segment growth rises from -0.26 percent to 4.19 percent.
- Peripheral segment growth rises from -3.70 percent to 2.05 percent.

Interpretation: Productivity effects are not just statistically significant; they are economically meaningful. The range between low- and high-productivity segments can shift predicted growth by several percentage points.

Economic message: The model has practical bite: relative productivity meaningfully predicts where firms expand and where they contract.

Table IV

Segment Growth

Regressions test the effects of plant-level productivity and industry-level demand on firm industry segment sales growth for main and peripheral segment of multiple-segment firms. The dependent variable, firm segment growth is industry adjusted in each year. Segment size and productivity are industry adjusted in each year. Real industry shipments is detrended by regressing industry shipments (in 1987 dollars) on a yearly time trend. Multiple-segment firms are firms with at least two segments each with 10 percent of their total shipments. Data are aggregated into firm three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber–White. Data are yearly from 1979 to 1992. Observations are included for growth rates less than 100 percent. (p-values are in parentheses.)

Dependent Variable: Segment Growth			
	Multiple-segment Firms		Test for Significant Diff.: Multiple-segment Interaction Variable (p-value)[d]
Variable	Main Segments	Peripheral Segments	
Constant	0.010	−0.016	
	(0.000)[a]	(0.000)[a]	(0.000)[a]
Real industry shipments (detrended)	−0.009	0.010	
	(0.408)	(0.271)	(0.155)
Firm segment productivity (TFP)[e]	0.046	0.058	
	(0.000)[a]	(0.000)[a]	(0.000)[a]
Industry shipments (detrended) * TFP	0.014	0.022	
	(0.292)	(0.019)[b]	(0.556)
ln(lagged firm segment size)	−0.012	−0.001	
	(0.149)	(0.511)	(0.178)
Number of plants owned by firm (beginning of year, coeff * 1000)	−0.418	−0.182	
	(0.001)[a]	(0.000)[a]	(0.933)
Firm multiple-segment variables			
Other segment's weighted TFP[f]	−0.006	−0.005	
	(0.030)[b]	(0.089)[c]	(0.928)
Relative demand * other segments weighted TFP[g]	−0.001	−0.006	
	(0.605)	(0.014)[b]	(0.178)
Total industry-segment years	32,517	55,398	
Chi-squared statistic	559.55	1474.38	
Significance level (p-value)	0.010	<1%	

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

[d] Significance test for a multiple-segment dummy variable interacted with each independent variable in a regression with all firm segments.

[e] Total Factor Productivity (TFP) is calculated using a translog production function.

[f] Other segments' productivity is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.

[g] Relative industry demand is interacted with other segments' productivity and equals one

Table IV: Segment growth for main and peripheral segments

Figure 1: Optimal production with differing ability across industries

Table V

Economic Significance of Regression Results

Predicted real growth rates of conglomerate and single-segment firms as productivity and change in shipments * size varies from the 25th to the 75th percentiles. We compute these predicted growth rates holding all variables except productivity and change in industry shipments * size at their sample medians. Predicted real growth rates using coefficient estimates from Tables III and IV: Variables taken at the respective sample medians.

	Growth Rate at the		
Varying Total Factor Productivity	25th Percentile	50th Percentile	75th Percentile
Single-segment firms (own productivity data):	−2.69%	0.92%	4.39%
Data from conglomerates' main segments (Both use coefficients from Table IV, column 1)	−2.17%	1.05%	4.36%
Conglomerate firms			
All segments (using coefficients from Table IV, column 2)	−2.34%	0.28%	2.97%
Main segments (using coefficients from Table V, column 1)	−0.26%	1.88%	4.19%
Peripheral segments (own productivity data)	−3.70%	−0.85%	2.05%
Productivity data from conglomerates' main segments (Both use coefficients from Table V, column 2)	−2.66%	0.10%	2.93%

Table V: Economic significance of regression results

6.9 Table VI: Segment capital expenditures

Read:

- Capital expenditures are positively related to segment productivity for single-segment firms, main segments, and peripheral segments.
- Industry shipments interacted with TFP is positive and significant in most columns.
- The relative-demand interaction with other segments' TFP is negative and significant for peripheral segments.
- These patterns mirror the segment-growth regressions.

Interpretation: The investment regressions show that the growth results reflect real capital allocation, not merely output fluctuations. Firms invest more in productive segments when industry conditions are favorable.

Economic message: Conglomerates appear to allocate investment toward productive opportunities and away from segments whose internal opportunity cost rises because other segments are more attractive.

Table VI

Segment Capital Expenditures

Regressions test the effects of plant-level productivity and industry-level demand on a firm's capital expenditures in a segment for single-segment and multiple-segment firms. The dependent variable, segment capital expenditures, is divided by the beginning of period capital stock and is industry adjusted in each year. Productivity and segment size are also industry adjusted in each year. Real industry shipments is detrended by regressing industry shipments (in 1982 dollars) on a yearly time trend. Data are aggregated into firm three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber–White. Data are yearly from 1979 to 1992. Observations are included for firms whose real value of shipments exceeds one million dollars and when observations are less than 100 percent in absolute value. (*p*-values are in parentheses.)

Dependent Variable: Segment Capital Expenditures				Test for Significant Diff.: Across Conglomerates Column 2–3 (*p*-value)[d]
		Multiple-Segment Firms		
Variable	Single-segment Firms	Main Segments	Peripheral Segments	
Constant	−0.033	0.001	−0.044	
	(0.000)[a]	(0.682)	(0.000)[a]	(0.000)[a]
Real industry shipments (detrended)	−0.001	−0.003	−0.001	
	(0.788)	(0.774)	(0.908)	(0.880)
Firm segment productivity (TFP)[e]	0.160	0.111	0.095	
(coefficient * 10)	(0.000)[a]	(0.000)[a]	(0.000)[a]	(0.054)[c]
Industry shipments (detrended) * TFP	0.016	0.033	0.013	
	(0.000)[a]	(0.018)[b]	(0.019)[b]	(0.725)
ln(lagged firm segment size)	−0.005	−0.001	−0.001	
	(0.000)[a]	(0.324)	(0.136)	(0.816)
Number of plants owned by firm	0.003	0.002	0.000	
(beginning of year, coeff * 10)	(0.227)	(0.265)	(0.092)[c]	(0.749)
Firm multiple-segment variables				
Other segments weighted TFP[f]		−0.001	−0.001	
		(0.220)	(0.229)	(0.554)
Relative demand * other segments		−0.004	−0.003	
weighted TFP[g]		(0.094)[c]	(0.038)[b]	(0.724)
Total industry-segment years	176,689	36,806	60,772	
Chi-squared statistic	810.49	37.62	134.88	
Significance level (*p*-value)	<1%	<1%	<1%	

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.
[d] Significance test for a multiple-segment dummy variable interacted with each independent variable in a regression with all firm segments.
[e] Total Factor Productivity (TFP) is calculated using a translog production function.
[f] Other segments' productivity is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.
[g] Relative industry demand is interacted with other segments' productivity and equals one

Table VI: Segment capital expenditures

Table VII

Growth and Alternative Measures of Efficiency

Regressions test the effects of plant-level value added per worker and cash flow on firm industry segment sales growth for multiple-segment firms. The dependent variable, firm segment growth, is industry adjusted in each year. Value added per worker and operating margin are also industry adjusted in each year. Value added is sales less materials cost of good sold divided by the number of workers. Operating margin is sales less materials cost of goods sold, employee costs, and capital expenditures divided by sales. Real industry shipments is detrended by regressing industry shipments (in 1982 dollars) on a yearly time trend. Multiple-segment firms are firms with at least two segments each with 10 percent of their total shipments. Data are aggregated into firm three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for fixed effects and correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors. Data are yearly from 1979 to 1992. Observations are included for growth rates less than 100 percent. (*p*-values are in parentheses.)

Dependent Variable: Segment Growth		
	Multiple-segment Firms	
Variable	Main Segments	Peripheral Segments
Constant	0.008	−0.018
	(0.000)[a]	(0.046)[a]
Real industry shipments (detrended)	0.005	0.009
	(0.614)	(0.261)
Efficiency measure		
Value added per worker	0.180	
	(0.000)[a]	
Operating margin		0.147
		(0.000)[a]
Industry shipments (detrended) * Efficiency measure	0.002	0.467
	(0.032)[b]	(0.000)[b]
ln(lagged firm segment size)	−0.006	−0.007
	(0.137)	(0.065)[c]
Number of plants owned by firm (beginning of year, coeff * 1000)	−0.693	−0.669
	(0.000)[a]	(0.000)[a]
Firm multiple-segment variables		
Other segment's weighted efficiency[d]	0.009	−0.015
	(0.323)	(0.213)
Relative demand * other segments weighted efficiency[e]	−0.010	−0.073
	(0.034)[b]	(0.000)[a]
Total industry-segment years	87,915	87,915
Chi-squared statistic	187.18	46.52
Significance level (*p*-value)	<1%	<1%

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.
[d] Other segments' efficiency is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.
[e] Relative industry demand is interacted with other segments' efficiency and equals one (zero,

Table VII: Growth and alternative measures of efficiency

6.10 Table VII: Growth and alternative measures of efficiency

Read:

- The paper replaces baseline TFP with value added per worker and operating margin.
- Efficiency measures remain positively related to growth.
- The interaction between industry shipments and efficiency is positive.
- The interaction between relative demand and other segments' weighted efficiency is negative.

Interpretation: The main results are not an artifact of the translog TFP measure. Alternative efficiency proxies support the same allocation logic.

Economic message: The evidence for efficient allocation is robust to different ways of measuring segment efficiency.

6.11 Table VIII: Extent of conglomerate diversification

Read:

- The table tests whether focus, measured by a sales Herfindahl, changes growth and investment sensitivity to productivity.
- Productivity and the industry-shipments-by-productivity interaction remain important.
- Herfindahl interactions are not uniformly significant.
- The key negative interaction between relative demand and other-segment TFP remains present.

Interpretation: The extent of diversification matters less than segment productivity and relative demand. Focus alone is not the central determinant of whether resources move efficiently.

Economic message: This result cautions against treating diversification itself as the mechanism. The allocation problem depends on comparative advantage across segments.

6.12 Table IX: Conglomerate firms that restructure

Read:

- Conglomerates that later restructure do not show the same strong model-consistent interaction patterns before restructuring.
- In contrast, conglomerates with no or limited restructuring show significant positive own TFP-demand interactions and negative other-segment interactions.
- The no-restructuring group has 70,787 industry-segment years, making it the main population of surviving conglomerates.

Interpretation: The table identifies a subset of conglomerates that may have behaved inefficiently or faced organizational problems. Importantly, this does not characterize the majority of conglomerates.

Economic message: The paper's conclusion is nuanced: agency problems can exist, but they are not the dominant explanation for the average resource allocation patterns in the data.

Table VIII

Extent of Conglomerate Diversification

Regressions test whether the extent of diversification influences segment resource allocation for multiple-segment firms. The extent of diversification is captured by a firm "Herfindahl." This Herfindahl is the sum of squared segment sales divided by total firm sales. The dependent variable in column one is firm segment growth while the dependent variable in column two is segment capital expenditures scaled by beginning of period capital stock. Real industry shipments is detrended by regressing industry shipments (in 1982 dollars) on a yearly time trend. Industry averages are subtracted in each year for both dependent variables. Segment size and productivity are also industry adjusted in each year. Data are aggregated into firm three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors. Observations are included for growth rates less than 100 percent and capital expenditures less than the beginning of period capital stock. Regressions are limited to firms whose real value of shipments is greater than 10 million dollars. Data are yearly from 1979 to 1992. (p-values are in parentheses.)

Dependent Variable: Segment Growth and Capital Expenditures

	All Divisions of Multiple-Segment Firms with Herfindahl Measured at the			
	2-Digit SIC Code		3-Digit SIC Code	
Variable	Segment Growth	Capital Expenditures	Segment Growth	Capital Expenditures
Constant	−0.004	−0.038	−0.004	−0.038
	(0.018)[b]	(0.000)[a]	(0.014)[b]	(0.000)[a]
Firm segment productivity (TFP)[d]	0.055	0.007	0.050	0.005
	(0.000)[a]	(0.000)[a]	(0.000)[a]	(0.001)[a]
Herfindahl * TFP	−0.003	0.003	0.007	0.009
	(0.603)	(0.293)	(0.220)	(0.013)[b]
Industry shipments (detrended)	0.003	−0.006	0.003	−0.006
	(0.629)	(0.137)	(0.615)	(0.130)
Industry shipments * TFP	0.020	0.011	0.020	0.011
	(0.008)[a]	(0.022)[b]	(0.008)[a]	(0.022)[b]
ln(lagged firm segment size)	−0.002	−0.184	−0.002	−0.184
(coefficient * 100)	(0.387)	(0.000)[a]	(0.379)	(0.000)[a]
Number of plants owned by firm	−0.285	0.025	−0.284	0.028
(beginning of year, coeff * 1000)	(0.000)[a]	(0.305)	(0.000)[a]	(0.247)
Firm Multiple-Segment Variables				
Other segment's weighted TFP[e]	−0.008	0.005	0.004	0.003
	(0.163)	(0.203)	(0.433)	(0.349)
Herfindahl * other segments TFP	0.001	−0.003	−0.006	−0.002
	(0.909)	(0.537)	(0.572)	(0.791)
Relative demand * other segments	−0.004	−0.002	−0.004	−0.002
weighted TFP[f]	(0.013)[b]	(0.077)[c]	(0.013)[b]	(0.077)[c]
Total industry-segment years	86,812	89,939	86,812	89,939
Chi-squared statistic	2042.74	164.09	2044.23	168.82
Significance level (p-value)	<1%	<1%	<1%	<1%

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

[d] Total Factor Productivity (TFP) is calculated using a translog production function.

[e] Other segments' productivity is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.

[f] Relative industry demand is interacted with other segments' productivity and equals one (zero, minus one)

Table VIII: Extent of conglomerate diversification

Table IX

Conglomerate Firms that Restructure

Regressions test the effects of plant-level productivity and industry-level demand on firm industry segment sales growth. We separate out firms that have had major restructuring over the period. Firms are classified as having undergone major restructuring if they decrease the number of segments they operate by more than 25 percent by the last year they are in the database. The dependent variable, firm segment growth, is industry adjusted in each year. Real industry shipments is detrended by regressing industry shipments (in 1982 dollars) on a yearly time trend. Segment size and productivity are also industry adjusted in each year. Data are aggregated into three-digit SIC codes for industry segments from underlying plant-level data. We estimate the regressions using unbalanced panel regressions allowing for correlated residuals within panel units. Significance tests are conducted using heteroskedasticity-consistent standard errors following Huber–White. Data are yearly from 1979 to 1992. Observations are included for growth rates less than 100 percent and for firms whose real value of shipments is greater than one million dollars. (p-values are in parentheses.)

Dependent Variable: Firm Industry Segment Growth

	Conglomerate Firms that had Major Restructuring		Conglomerate Firms with No or Limited Restructuring
	With 2–3 Segments	With >3 Segments	
Variable	(Years Before Restructuring)		(All Years)
Constant	0.010	−0.020	−0.001
	(0.174)	(0.000)[a]	(0.505)
Real industry shipments (detrended)	0.056	0.015	−0.001
	(0.028)[b]	(0.343)	(0.943)
Firm segment productivity (TFP)[d]	0.064	0.064	0.052
	(0.000)[a]	(0.000)[a]	(0.000)[a]
Industry shipments (detrended) * TFP	−0.048	−0.017	0.026
	(0.115)	(0.411)	(0.003)[a]
ln(lagged firm segment size)	0.015	−0.013	−0.001
	(0.834)	(0.215)	(0.741)
Number of plants owned by firm	−4.939	−0.073	−0.346
(beginning of year, coeff * 1000)	(0.000)[a]	(0.238)	(0.000)[a]
Firm Multiple-Segment Variables			
Other segment's weighted TFP[e]	−0.005	−0.002	−0.008
	(0.370)	(0.764)	(0.001)[a]
Relative demand * other segments	0.003	0.005	−0.005
weighted TFP[f]	(0.553)	(0.393)	(0.012)[b]
Total industry segment years	6,259	10,648	70,787
Total firms	1,075	510	3,629
Chi-squared statistic	231.93	341.96	1605.62
Significance level (p-value)	<1%	<1%	<1%

[a,b,c] Significantly different from 0 at the 1 percent, 5 percent, and 10 percent levels, respectively, using a two-tailed test.

[d] Total Factor Productivity (TFP) is calculated using a translog production function.

[e] Other segments' productivity is a weighted average of the firm's other segment(s) weighted by the segment(s) predicted output.

Table IX: Conglomerate firms that restructure

7 Interpretive synthesis

- **Conglomerate discount versus inefficiency.** The paper separates a valuation fact from a mechanism. A discount can arise because conglomerates are less productive, not because they misallocate resources.
- **Comparative advantage inside firms.** The largest segments are more productive, and productive segments respond more strongly to demand shocks. This fits an optimizing model in which firms expand where they have skill.
- **Other-segment opportunity cost.** A focal segment's growth falls when the firm's other segments become more attractive. This is the hallmark of an internal allocation model, not simple subsidization.
- **Heterogeneity among conglomerates.** Surviving conglomerates generally look efficient, while firms that later restructure look less model-consistent. The paper therefore does not deny agency problems; it limits their scope.
- **Methodological lesson.** Corporate finance tests of internal capital markets need segment-specific productivity measures. Industry-level q alone can confound investment opportunities with firm-specific ability.

8 Short summary

- The paper develops a neoclassical model of optimal diversification based on industry-specific organizational ability and decreasing returns to firm scale.
- Diversification is optimal when the firm's abilities across industries are sufficiently similar.
- The model predicts that larger segments should be more productive and that segments should grow in response to own productivity, own demand, and other-segment opportunities.
- The paper uses U.S. Census plant-level manufacturing data from 1974–1992 and constructs firm-industry segments by three-digit SIC code.
- Productivity is estimated using translog production functions with plant-firm fixed effects.
- Conglomerates are generally less productive than single-segment firms of similar size, consistent with an endogenous conglomerate discount.
- Within conglomerates, main segments are more productive than peripheral segments.
- Segment growth and investment are positively related to segment productivity and to productivity interacted with industry demand.
- Segments grow less when other segments have stronger productivity and demand opportunities.
- Alternative efficiency measures and diversification Herfindahl tests support the main results.
- Conglomerates that later restructure fit the model less well, suggesting some agency problems in a subset of firms.
- The majority of conglomerates appear to allocate resources consistently with efficient comparative-advantage behavior.

9 Conclusion

9.1 Core conclusion

Maksimovic and Phillips argue that conglomerate resource allocation should be evaluated against a productivity-based neoclassical benchmark. Using plant-level data, they find that the majority of conglomerates allocate resources in a manner consistent with efficient growth across industries, not with systematic cross-subsidization of weak segments.

9.2 Economic intuition

Firms have different organizational ability across industries. A firm expands in the segment where its productivity and industry opportunities are strongest, but because managerial and organizational capacity is scarce, growth in one segment raises the opportunity cost of growth elsewhere. A conglomerate discount can therefore reflect equilibrium sorting and lower productivity rather than inefficient internal capital markets.

9.3 Incremental contribution takeaway

The paper's distinctive contribution is to show that a micro-founded productivity benchmark can reverse the interpretation of conglomerate investment patterns. Once plant-level productivity is measured directly, many patterns previously read as evidence of inefficient internal capital markets become consistent with profit maximization and comparative advantage.

9.4 Limits and final takeaway

The paper does not rule out overhead waste, managerial private benefits, or agency problems in all conglomerates. It does show that these explanations are not necessary to explain the average segment growth and investment patterns. The final takeaway is that diversification can destroy measured value without necessarily destroying resources through inefficient allocation.